Coding Exercise

For this exercise we would like to see how you go about solving a rather straight forward coding challenge and architecting for the future. One of our key values in how we develop new systems is to start with very simple implementations and progressively make them more capable, scalable and reliable. And releasing them each step of the way.

As you work through this exercise it would be good to "release" frequent updates by pushing updates to a shared git repo (we like to use Gitlab's free private repos for this — add jen.coopersmith as a project member). It's up to you how frequently you do this and what you decide to include in each push.

Here's what we would like you to build:

URL lookup service

We have an HTTP proxy that is scanning traffic looking for malware URL's. Before allowing HTTP connections to be made, this proxy asks a service that maintains several databases of malware URL's if the resource being requested is known to contain malware. Write a small web service, in the language/framework of your choice, that responds to GET requests where the caller passes in a URL and the service responds with some information about that URL. The GET requests look like this:

```
GET /urlinfo/1/{hostname_and_port}/{original_path_and_query_string}
```

The caller wants to know if it is safe to access that URL or not. As the implementor you get to choose the response format and structure. These lookups are blocking users from accessing the URL until the caller receives a response from your service, so make sure it’s performant.

Your finished product should include:

- necessary documentation for us to set up the service ourselves on EC2/Vagrant/Docker or similar system of your choosing. Do not utilize services such as AppEngine or Heroku. Bonus if you do this using Ansible/Salt/Chef or other configuration management.
- documentation on your solution’s result format/structure
- ability to handle an expected load of thousands of requests per second
- unit tests. Bonus if you have integration tests as well, but not required.

Give some thought to the following:

- The size of the URL list could grow infinitely, how might you scale this beyond the memory capacity of your single host? Bonus if you implement this.
- The number of requests may exceed the capacity of your initial solution, how might you scale beyond? Bonus if you implement this.
- What are some strategies you might use to update the service with new URLs? Updates may be as much as 5 thousand URLs a day with updates arriving every 10 minutes.